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Mitigating Proppant Flowback in Area-1 Fields: A Case Study on the Implementation of Rod-Shaped Proppant

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Abstract

Proppant flowback has been a significant issue following hydraulic fracturing operations in several fields within the Area-1. It adversely impacts both surface and downhole equipment, leading to reduced production rates. Various methodologies to minimize this problem have been considered. However, the reservoir properties pose challenges for the implementation of some technologies. To mitigate this issue, recent hydraulic fracturing treatments in select wells have utilized rod-shaped proppant. This proppant not only interlocks to minimize flowback but also offers higher porosity compared to spherical grains.

The trial was planned and executed in three steps: pumpability of the material, proof of concept, and optimization treatment. In the first step, the focus was on ensuring the material could be pumped effectively, meeting both surface equipment and reservoir requirements. During the proof-of-concept phase, the objective was to assess the reduction in proppant flowback before proceeding to optimize the treatment. A thorough evaluation of similar treatments worldwide was conducted to minimize potential issues. Post-fracturing surveillance was also performed to evaluate the results of the treatment.

Despite learning from similar projects and implementing a "what to do" list, the first deployment encountered early job termination. Upon review, it was confirmed that the failure of the placement was caused by the limited width of the fracture and the minimal entrance hole of the perforation.

In the second trial, a revised approach and requirements successfully placed the rod-shaped proppant in several wells, confirming the material's place-ability. During this phase, production results from several wells were evaluated, confirming a reduction in proppant flowback with similar production parameters. A further long-term study is currently being conducted to evaluate the long-term impact on production and proppant flowback.

This paper provides a comprehensive background on the utilization of technology. It also outlines the step-by-step implementation process, and the lessons learned throughout. The paper highlights the challenges faced and the adjustments made to critical factors such as perforation strategy and fracture width. The insights gained from this study offer valuable guidance for future technological deployment in similar fields.

Introduction

Hydraulic fracturing (HF) operations have been successfully implemented in the Area-1, operated by Medco Oman. Discovered in 1991, the Area-1 cluster is located in southern Oman on the eastern flank of the South Oman Salt Basin—a geologically complex region with significant hydrocarbon potential (see Figure 1).



Figure 1—Map of the Area-1 Location in Oman

The field contains hydrocarbons in multiple formations, including the Gharid, Haradh, Karim, and the deeper Huqf carbonate layers. These accumulations are structurally controlled, primarily forming within faulted and domal anticlines that act as effective traps for oil.

To date, over 50 hydraulic fracturing jobs have been executed across several fields within the Area-1 cluster, including Simsim, Waradh, Zhara, Amin South, and Jadeer. The primary objective of these campaigns has been to economically enhance production by improving reservoir connectivity in low-permeability formations. This is achieved by creating high-conductivity fractured networks that facilitate the efficient flow of hydrocarbons to the wellbore.

While these operations have generally succeeded in boosting production, a recurring challenge has been proppant flowback—the return of proppant material to the surface during post-fracturing cleanup and production. This issue can lead to:

- Reduced production efficiency
- Damage to surface and downhole equipment
- Increased operational costs due to required well interventions

Proppant flowback typically occurs during the cleanup phase, which can last from several days to weeks, but in some cases, it may persist throughout the well's productive life. To address this, further optimization of proppant selection and placement techniques is essential to ensure long-term well integrity and sustained production.

Several industry-proven techniques have been employed to mitigate proppant flowback (Edelman et al., 2013), including:

- Resin-Coated Proppants (RCP)
- Force Closure Techniques
- Fiber-based Technologies

Among the three available technologies in the market designed to mitigate proppant flowback, rod-shaped proppants have been selected for several compelling reasons. Research by L. Sortino et al. (2010) highlights the effectiveness of rod-shaped proppants in enhancing production in low-permeability reservoirs. These proppants offer improved mechanical stability and embedment resistance. Greg McDaniel et al. (2010) further emphasized the role of elongated, high-strength particles in maintaining fracture conductivity and controlling proppant flowback.

In response to the challenges faced in earlier campaigns, Medco Oman has adopted a novel approach in its latest HF operations by deploying rod-shaped proppants. This technology is expected to; significantly reduce proppant flowback, enhance fracture conductivity and improve long-term production performance.

Additionally, the campaign will incorporate advanced monitoring and control systems to track proppant behavior in real time and proactively manage flowback risks. These systems will support data-driven decision-making and continuous optimization of HF operations.

Rod-Shaped Proppant Technology

The rod-shaped proppant is a specialized type of proppant characterized by its elongated, high-strength particle design. It typically measures between 0.10 to 0.11 inches in length and 0.05 to 0.06 inches in diameter, as illustrated in Figure 2 below.



Figure 2—Appearance of rod-shaped proppant.

The effectiveness of rod-shaped proppants in resisting flowback is primarily attributed to mechanical interference rather than chemical bonding. These elongated particles interlock with one another, forming a

stable and consolidated structure that resists displacement. This mechanical stability is crucial for preventing the proppant from being carried away by the flow of fracturing fluids or hydrocarbons during production.

Due to their unique physical properties, rod-shaped proppants significantly improve proppant pack conductivity (Greg McDaniel et al., 2010). Their distinctive packing behavior leads to the formation of a more porous and stable structure, which enhances fluid flow through the fracture. As shown in Figure 3, the alignment of rod-shaped particles creates larger void spaces between them, resulting in increased porosity and higher resistance to the forces that typically cause proppant flowback.

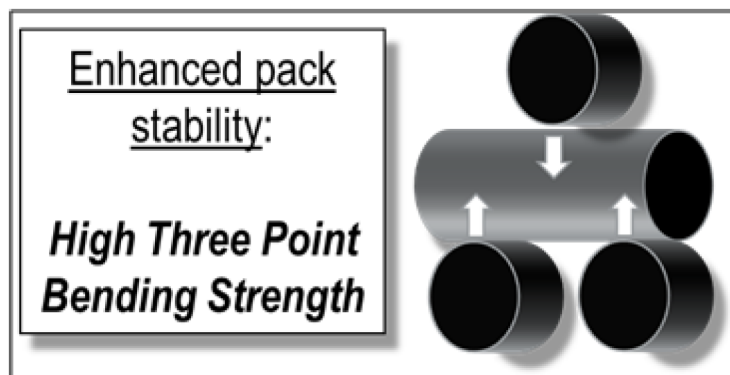


Figure 3—Three-point bending strength.

This improved conductivity not only supports better hydrocarbon flow but also facilitates a more efficient fracture cleanup process. By allowing fracturing fluids to be removed more effectively, the rod-shaped proppants contribute to a longer effective fracture half-length, assuming similar proppant placement and reservoir conditions.

Rod-Shaped Proppant Field Trial

Four wells have been selected for the trial. Given the limited prior application of this technology in the area, the trial will rely on real-time, on-the-fly decision-making to optimize the treatment and ensure the objectives are met. This adaptive approach allows the team to respond dynamically to downhole conditions and operational feedback, maximizing the effectiveness of the rod-shaped proppant deployment.

Pumpability of the Material Trial

As successfully implemented in other countries, we have begun evaluating the pumpability of rod-shaped proppant in one of the designated wells—Well A. The fracture design for this well includes a total of 110,000 lbs. of proppant, with 18% rod-shaped proppant incorporated as a tail-in at concentrations of 7 PPA and 8 PPA.

This approach is intended to ensure that the near-wellbore area is effectively packed with rod-shaped proppant, which plays a crucial role in preventing proppant flowback and maintaining fracture integrity (Figure 4).

#	Stage	Fluid	Prop.	Prop Con PPA	Pump Rate BPM	CF Vol bbl	Sl. Vol bbl	Prop., lb	Cum. Prop., lb	Time min	Cum. time min	J475 (ppt)	J218 (ppt)
1	Pad	YF130D			20	240.0	240.0	0	0	12.0	12.0	1.5	0
2	1 PPA	YF130D	20/40	1	20	80.0	83.0	3,360	3,360	4.2	16.2	1.5	0
3	2 PPA	YF130D	20/40	2	20	80.0	86.1	6,720	10,080	4.3	20.5	1.5	0
4	3 PPA	YF130D	20/40	3	20	50.0	55.7	6,300	16,380	2.8	23.2	2	0
5	3 PPA	YF130D	16/30	3	20	30.0	33.4	3,780	20,160	1.7	24.9	2	0
6	4 PPA	YF130D	16/30	4	20	80.0	92.2	13,440	33,600	4.6	29.5	2	0
7	5 PPA	YF130D	16/30	5	20	80.0	95.2	16,800	50,400	4.8	34.3	2	0
8	6 PPA	YF130D	16/30	6	20	80.0	98.3	20,160	70,560	4.9	39.2	2	0.3
9	7 PPA	YF130D	16/30	7	20	65.0	82.3	19,110	89,670	4.1	43.3	2	0.3
10	7 PPA	YF130D	RodProp	7	20	30.0	37.0	8,820	98,490	1.8	45.2	3	0.5
11	8 PPA	YF130D	RodProp	8	20	35.0	44.3	11,760	110,250	2.2	47.4	3	0.5
11	Flush*	WF130			20	57.8	57.8	0	110,250	2.9	50.3	0	2
12	Shut in									60.0	110.3		
TOTALS:						908	1,005	110,250		50.3		107.3	11.2
Pad %:						28.2%	25.3%						

Treating Fluids' Volumes (incl. DataFRAC)			
	bbl	gal	m3
Tr. Water	120.0	5,040	19.1
WF130	493.8	20,740	78.5
YF130D	1,000.0	42,000	159.0

Proppant Mass		
	lb	MT
BBSHield	5,040	2.3
20/40	16,380	7.4
16/30	73,290	33.2
RodProp	20,580	9.3

Pad Clean	28.2%
Pad Dirty	25.3%

Figure 4—Pumping Schedule for the 1st Trial of Rod-Shape Proppant

The pumping operation began with a Breakdown Test (Figure 5) to confirm the formation's breakdown pressure and to verify connectivity between the wellbore and the reservoir following perforation.

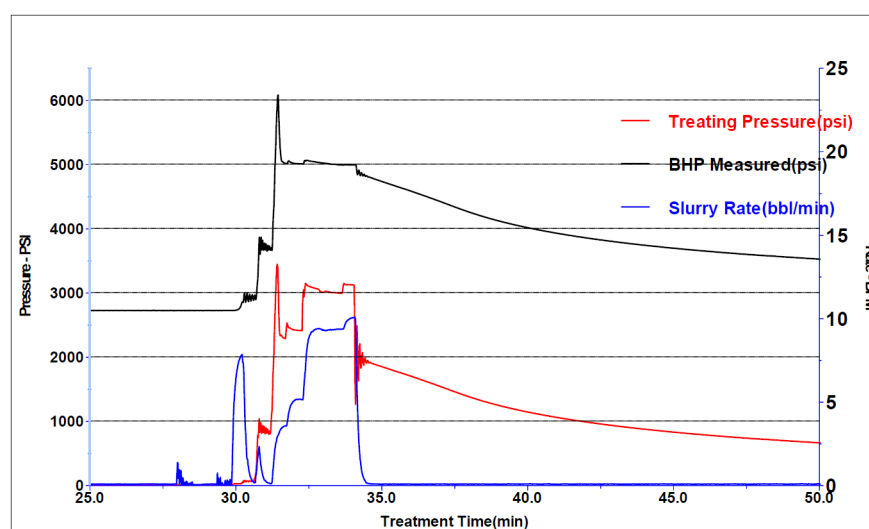


Figure 5—Breakdown test plot.

Following the breakdown, both a Step Rate Test and a Step-Down Test were conducted (Figure 6):

- The Step Rate Test was performed to determine the fracture extension pressure—the pressure at which the formation begins to propagate fractures.
- The Step-Down Test was used to estimate near-wellbore friction, providing insights into the resistance encountered by the fracturing fluid as it flows through the wellbore and perforations.

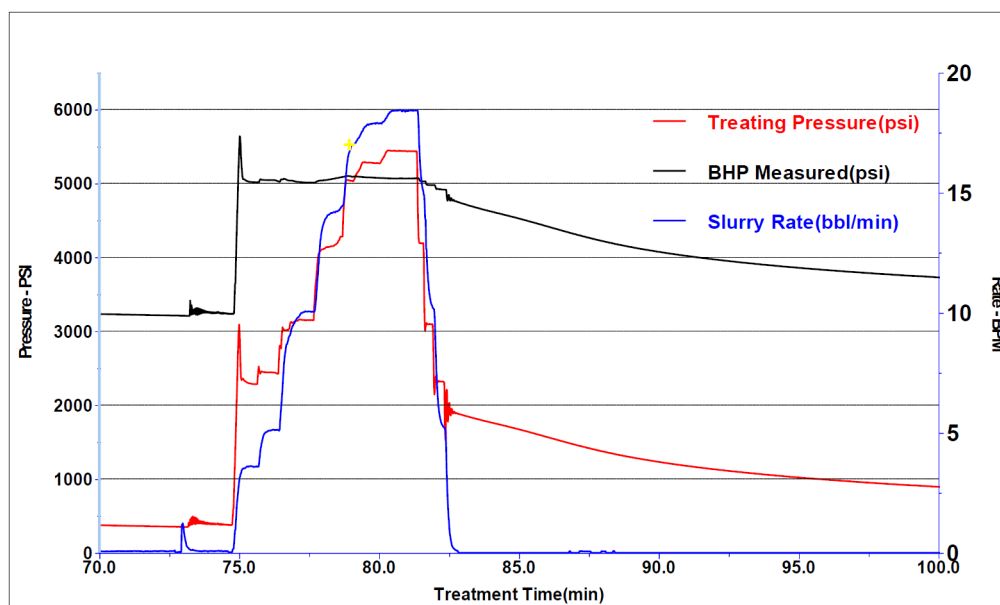


Figure 6—Step Rate Test and Step-Down Test Plot Well A

Subsequently, a DataFrac (Calibration Test) was conducted (Figure 7) to obtain key formation parameters:

- Closure Pressure (P_c): The pressure at which the induced fractures begin to close.
- Fluid Efficiency (FE): A measure of how effectively the fracturing fluid contributes to fracture creation and maintenance.

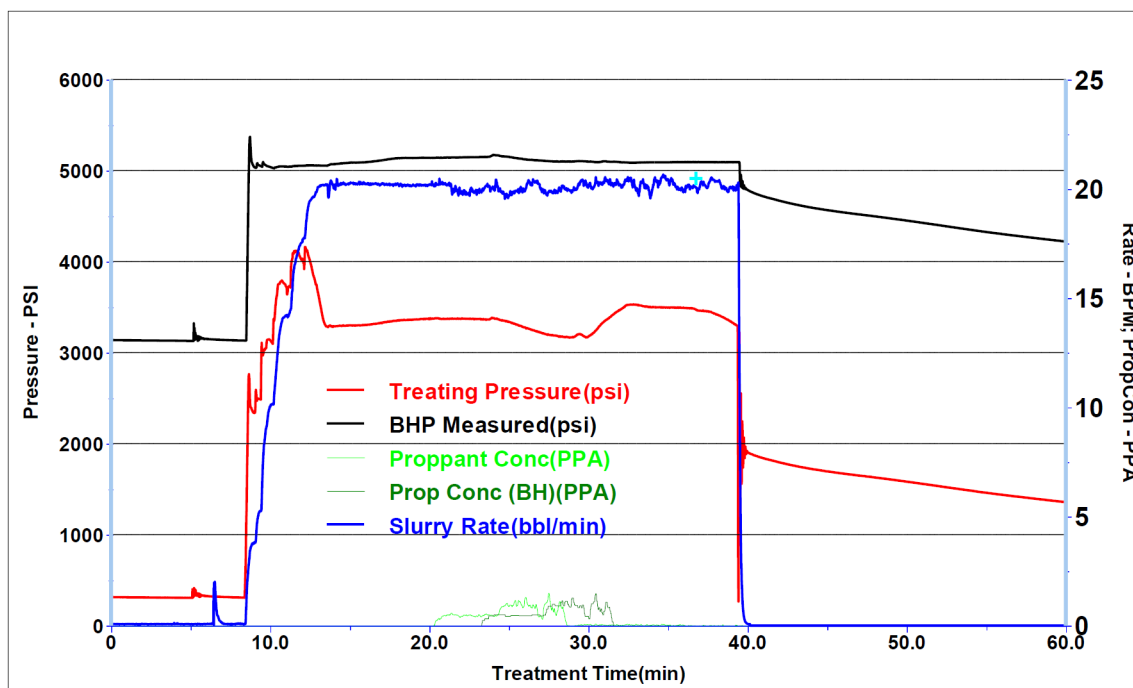


Figure 7—DataFrac / Calibration Test Plot Well A

Table-1 below presents the DataFrac result comparison with the offset well.

Table 1—DataFrac result comparison.

WELL	Vol Pumped (bbl)	Major Pumping Rate	ISIP	ISIP Grad (psi/ft)	Pc	Pc Grad (Psi/ft)	Pnet	Eff
W-42	140	18	4,540	0.70	3,958	0.61	582	30%
W-87	232	18	4,824	0.73	3,944	0.59	880	26%
W-95	121.2	18.6	4,578	0.69	3,912	0.59	666	35%
Well A	510	20.1	4,880	0.74	4,359	0.66	521	20%

In this DataFrac operation, a particulate pill was pumped at a low concentration of 0.5 PPA to 1 PPA to create an artificial stress barrier prior to the MainFrac treatment. The purpose of this barrier is to aid in fracture containment by preventing excessive downward height growth. By introducing this particulate pill, we aim to enhance the effectiveness of the MainFrac treatment by ensuring that the fracture remains within the desired vertical boundaries.

The results from the DataFrac test were utilized to calibrate the formation stress profile through pressure matching. Temperature Log surveillance was conducted after the DataFrac test to provide supporting information, indicating the height of the created fracture geometry. This calibration enabled a more accurate understanding of the in-situ stress distribution, which informed an updated fracture design, as illustrated in Figure 8. In this plot, Rod-shape proppant was designed to cover the near-wellbore area, as indicated by the color code of the specific proppant in the simulation.

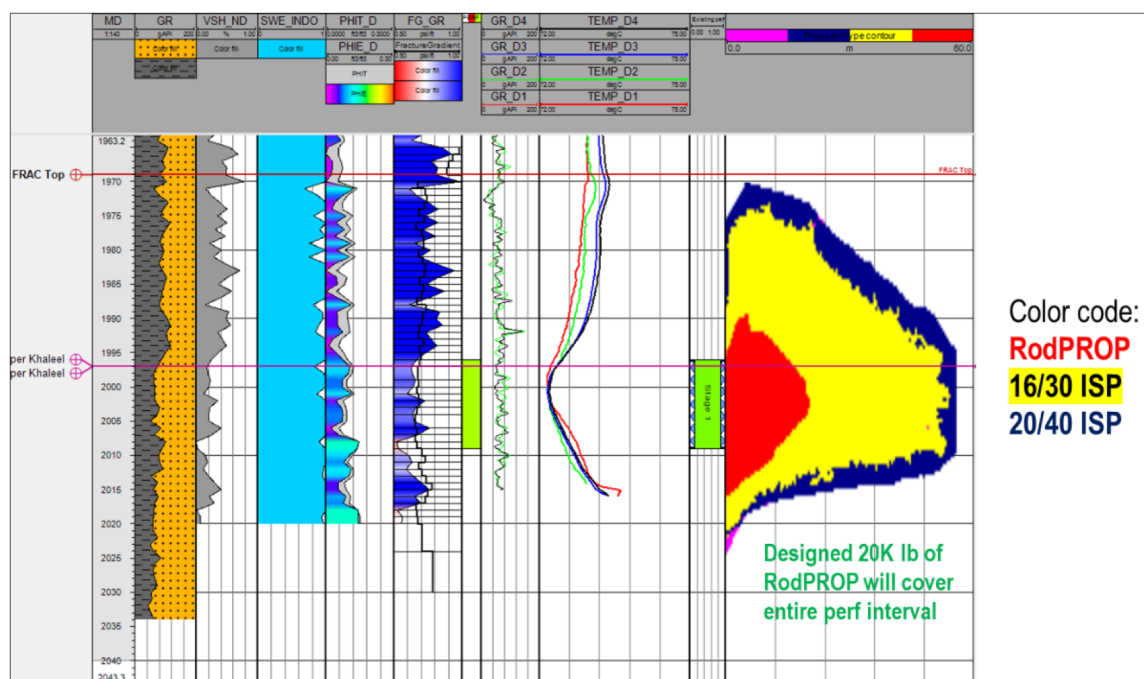


Figure 8—Frac re-design incorporates the Rod-shape proppant as tail in Well A

The main fracturing operation began smoothly, utilizing conventional 20/40 Intermediate Strength Proppant (ISP) and 16/30 ISP during the early stages, with no abnormal pressure trends observed. At a concentration of 7 PPA, the transition to rod-shaped proppant was initiated. During this transition, surface pressure trends remained stable at both 7 PPA and 8 PPA, indicating no immediate operational concerns.

However, once the 7 PPA rod-shaped proppant reached the perforation zone, a noticeable increase in pressure was observed. As a result, only 1,917 lbs of rod-shaped proppant were successfully placed into the formation before the operation was terminated early (Figure 9).

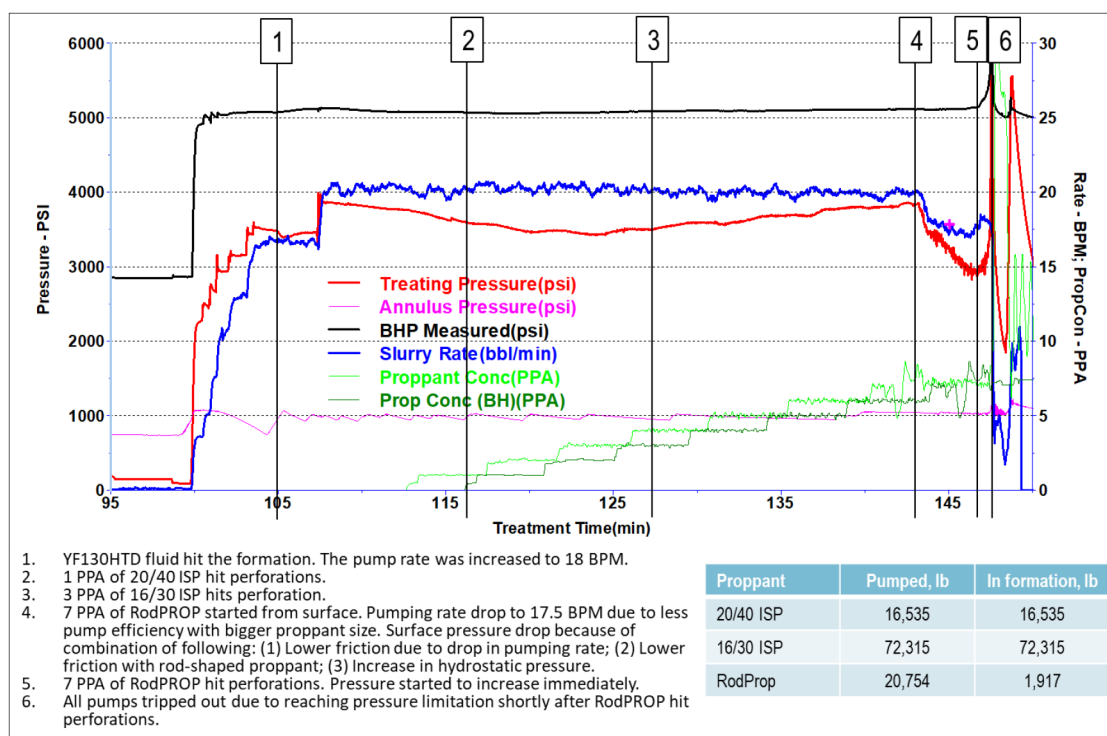


Figure 9—Main frac plot Well A

Pumping Optimization Trial

Following the first well trial—which resulted in a pressure-out event toward the end of the operation—the objective for the second well was to optimize the placement of rod-shaped proppant. Transitioning between different proppant sizes and types introduced operational challenges. However, best operational practices were established and strictly followed throughout the treatment.

When using rod-shaped proppant as a tail-in, it is common to observe fluctuations in proppant concentration during the transition phase. This is primarily due to the inevitable mixing of two proppants with different physical properties in the hopper. Additionally, a rapid drop in treatment pressure is often observed, accompanied by a notable reduction in friction pressure. This behavior is attributed to the elongated shape of the rod-shaped proppant, which tends to align with the flow direction, reducing turbulence and consequently lowering friction losses.

From the initial trial, it was observed that the formation stress in this well was significantly higher compared to offset wells in the same field. The treating pressure was approximately 700 psi higher than in other wells (Figure 10), despite being pumped at a similar pump rate, using the same fracturing fluid, and experiencing similar tubing friction. This indicates a higher stress environment due to the reservoir's tightness. This comparison is further illustrated in Figure 11.

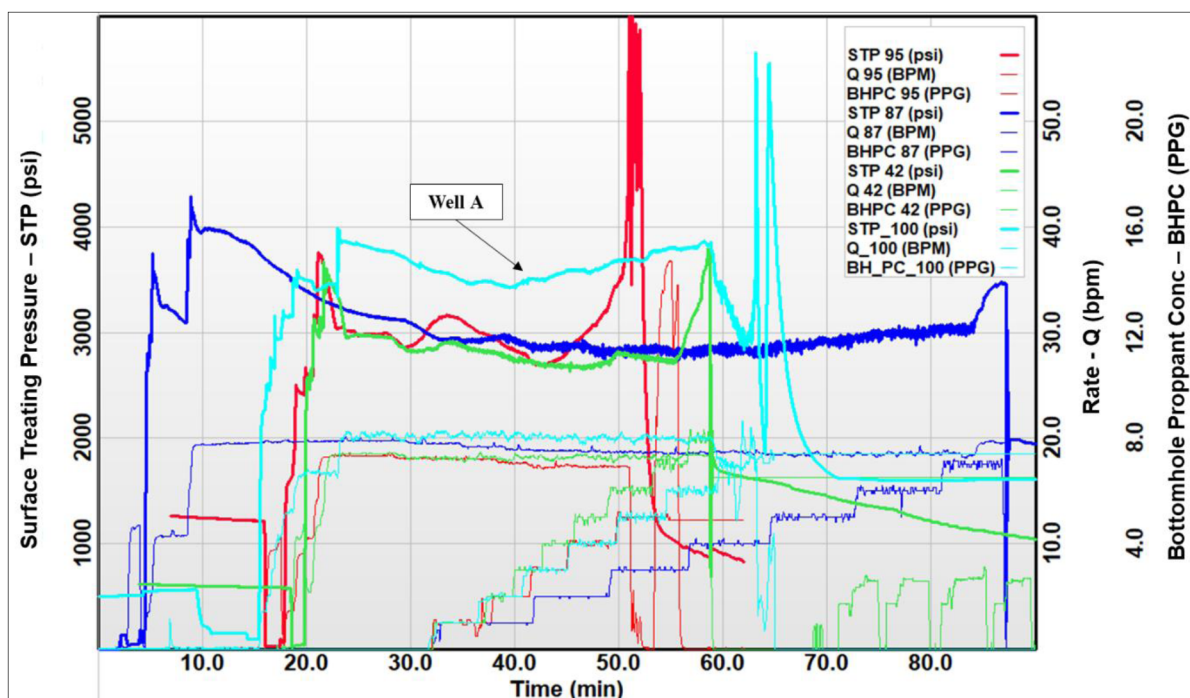


Figure 10—Main frac treating pressure comparison with offset wells.

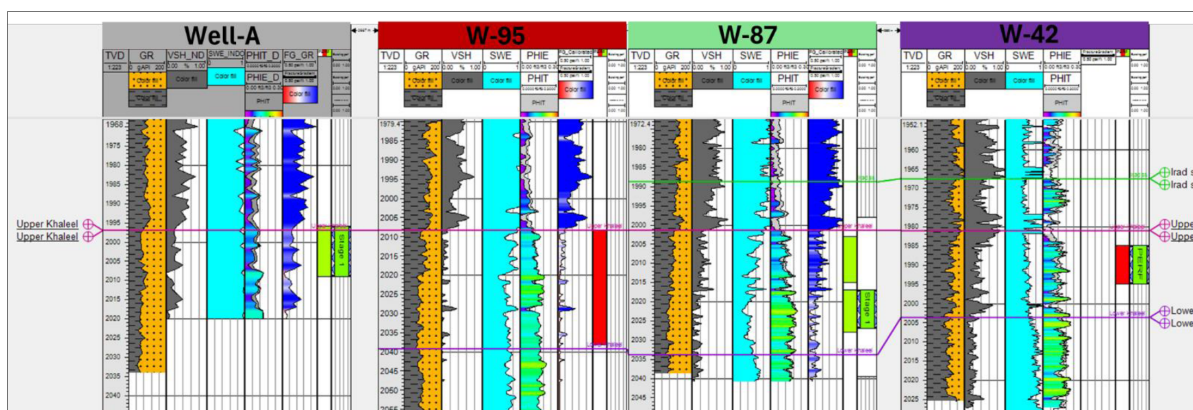


Figure 11—Stress profile comparison with offset wells.

These findings suggest that higher proppant concentrations may not be sufficient to enter the perforation tunnels. As a result, only a small amount of rod-shaped proppant was effectively placed inside the formation during the first trial.

The second trial was conducted in Well B, with several adjustments made to improve the outcome. Specifically, the rod-shaped proppant concentration was reduced to 6 PPA and 7 PPA. The operation adhered to the same best operational practices, and the stress profile was recalibrated using pressure matching supported by temperature log surveillance. Figure 12 illustrates the updated pumping schedule, which included a significant reduction in rod-shaped proppant volume—now comprising only 12% of the total proppant, placed as a tail-in during the final stage of the treatment.

#	Stage	Fluid	Prop.	Prop Con PPA	Pump Rate BPM	CF Vol bbl	Sl. Vol bbl	Prop., lb	Cum. Prop., lb	Time min	Cum. time min	J475 (ppt)	J218 (ppt)
1	Pad WF	WF130			18	100.0	100.0	0	0	5.6	5.6	1.5	0
2	BBShield	WF130	BBShield	0.5	18	50.0	51.0	1,050	1,050	2.8	8.4	1.5	0
3	BBShield	WF130	BBShield	1	18	50.0	51.9	2,100	3,150	2.9	11.3	1.5	0
4	Pad YF	YF130D			18	20.0	20.0	0	3,150	1.1	12.4	1.5	0.2
5	1 PPA	YF130D	20/40	1	18	60.0	62.3	2,520	5,670	3.5	15.8	1.5	0.2
6	2 PPA	YF130D	20/40	2	18	60.0	64.6	5,040	10,710	3.6	19.4	1.5	0.2
7	3 PPA	YF130D	20/40	3	18	40.0	44.6	5,040	15,750	2.5	21.9	2	0.5
8	3 PPA	YF130D	16/30	3	18	20.0	22.3	2,520	18,270	1.2	23.1	2	0.5
9	4 PPA	YF130D	16/30	4	18	70.0	80.7	11,760	30,030	4.5	27.6	2	0.5
10	5 PPA	YF130D	16/30	5	18	70.0	83.3	14,700	44,730	4.6	32.3	2	1
11	6 PPA	YF130D	16/30	6	18	50.0	61.4	12,600	57,330	3.4	35.7	2	1
12	6 PPA	YF130D	RodProp	6	18	20.0	24.0	5,040	62,370	1.3	37.0	2	1
13	7 PPA	YF130D	RodProp	7	18	10.0	12.3	2,940	65,310	0.7	37.7	2	1
14	Flush*	WF130			18	56.0	56.0	0	65,310	3.1	40.8	0	2
15	Shut in									60.0	100.8		
TOTALS:						676	734	65,310		40.8		52.5	14.9
Pad %:						35.5%	32.9%						

Treating Fluids' Volumes (Incl. DataFRAC)			
	bbl	gal	m3
Tr. Water	120.0	5,040	19.1
WF130	450.2	18,908	71.6
YF130D	420.0	17,640	66.8

Proppant Mass		
	lb	MT
BBShield	3,150	1.4
20/40	12,600	5.7
16/30	41,580	18.9
RodProp	7,980	3.6

Pad Clean	35.5%
Pad Dirty	32.9%

Figure 12—Optimized pumping schedule with less rod-shape proppant as a tail step.

Despite these adjustments, the second trial concluded with no rod-shaped proppant entering the formation, leading to the early termination of the operation (Figure 13).

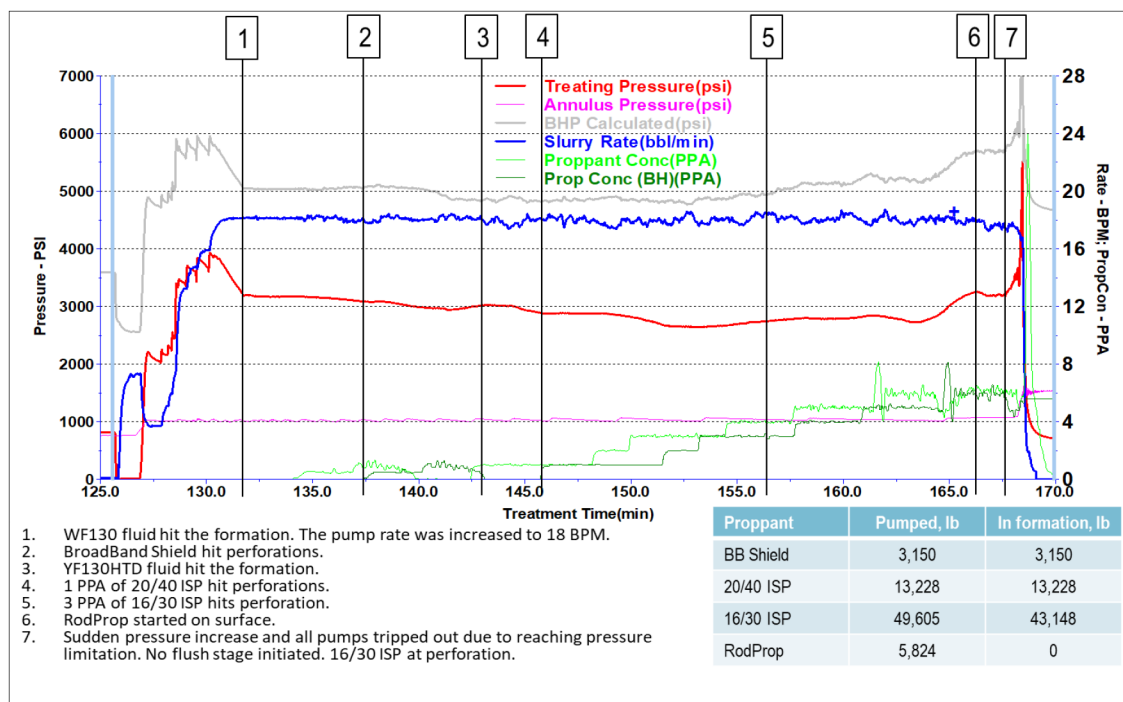


Figure 13—Main Frac Plot for 2nd trial of rod-shape proppant (Well B)

To address the challenges encountered in previous trials, the perforation strategy was revised by increasing the Entrance Hole Diameter (EHD) from 0.41" to 0.53". According to friction analysis conducted

in line with local best practices, this larger EHD is expected to reduce perforation friction, thereby facilitating the entry of rod-shaped proppant into the formation.

To achieve this larger EHD, the gun size was changed from 4-1/2" to 3-3/8". The detailed specifications of the perforation modifications are presented in [Table 2](#).

Table 2—Perforation detail changed from 4-1/2" to 3-3/8".

Gun Size	Charge	Shot Density (spf)	DOP (in)	EHD (in)	Area Flow (sqin/m)
4-1/2"	4539XP	5	45	0.41	2.16
3-3/8"	3324XS	6	22.7	0.53	4.34

Perforation Optimization Trial

Following the implementation of the new perforation strategy, a trial was conducted in Well C. The total volume of rod-shaped proppant was reduced to 5%, maintained at a concentration of 6 PPA. This approach aims to ensure adequate coverage of the near-wellbore area while minimizing operational risks. The trial was successfully executed, with approximately 6,600 lbs of rod-shaped proppant placed at 6 PPA without any issues ([Figure 14](#)).

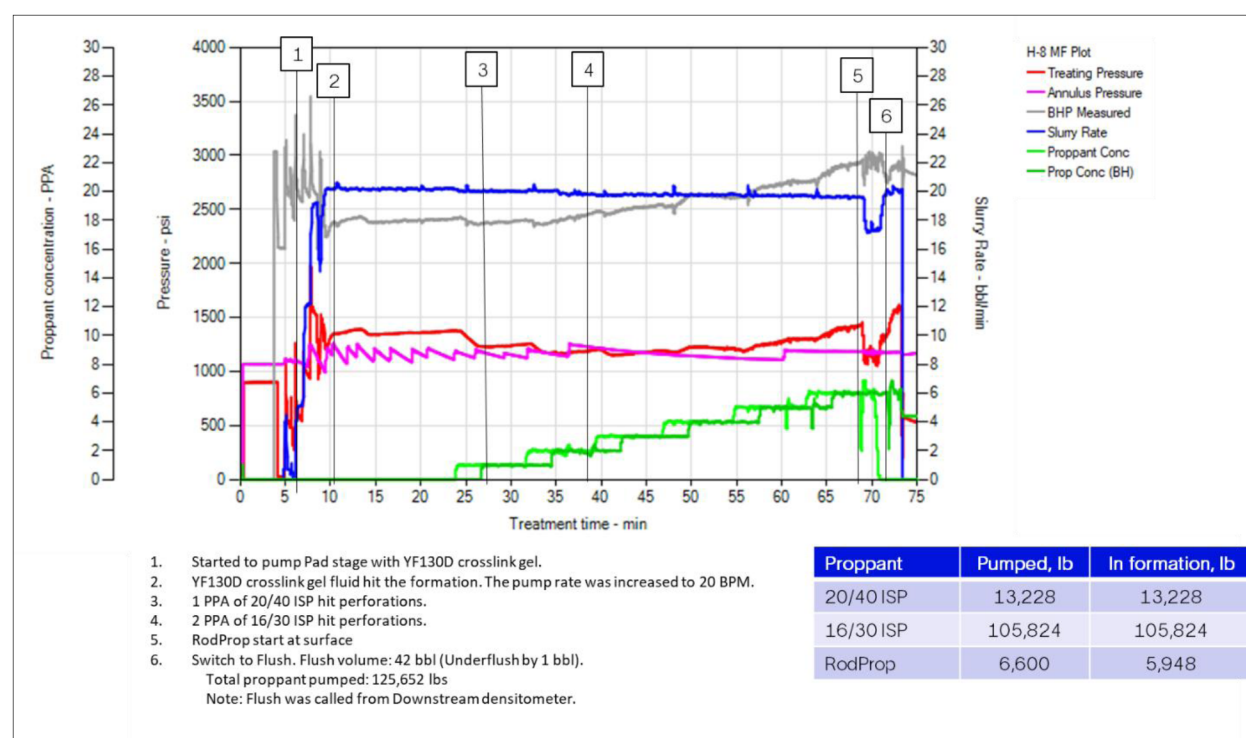


Figure 14—Main frac plot in Well C

Building on this success, another well D was selected for the next trial. This operation also concluded successfully, placing 6,600 lbs of rod-shaped proppant at a slightly higher concentration of 7 PPA ([Figure 15](#)).

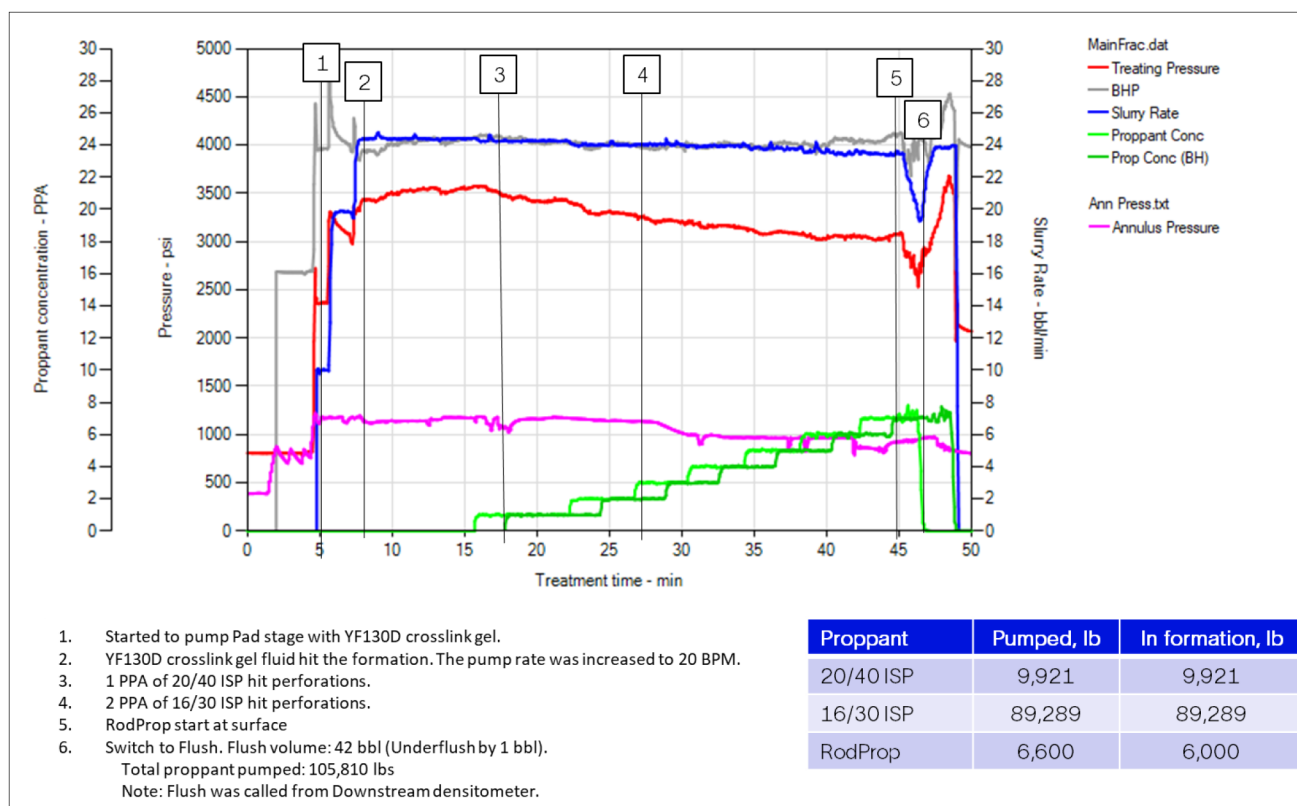


Figure 15—Main frac plot in Well D

Following the successful implementation in both wells, they were put into production, and their performance is being monitored over time to evaluate the long-term impact of the rod-shaped proppant placement.

During the monitoring period, both Well C and Well D demonstrated no signs of proppant flowback, and no well intervention was required after they were brought into production. In contrast, several offset wells used for comparison experienced operational issues that necessitated the use of a hoist (workover rig). These interventions were required to address problems such as parted rods in rod pumps, wellbore cleaning, and stuck rods.

This comparison is clearly illustrated in Figure 16, where Wells C and D show a longer operational life without any intervention, while the offset wells required a minimum of 2 to 4 hoist entries.

The absence of proppant flowback in Wells C and D can be attributed to the successful implementation of the revised perforation strategy and the optimized use of rod-shaped proppant. The increase in Entrance Hole Diameter (EHD) to 0.53" significantly improved proppant transport and placement, ensuring effective coverage of the near-wellbore area. This contributed to the stability and integrity of the proppant pack, preventing flowback and supporting sustained well productivity.

The long-term monitoring of these wells provided valuable insights into their performance and operational efficiency. The extended operational life of Wells C and D—achieved without the need for any well intervention—highlights the effectiveness of the new strategy. In contrast, the offset wells experienced frequent issues that required multiple hoist interventions.

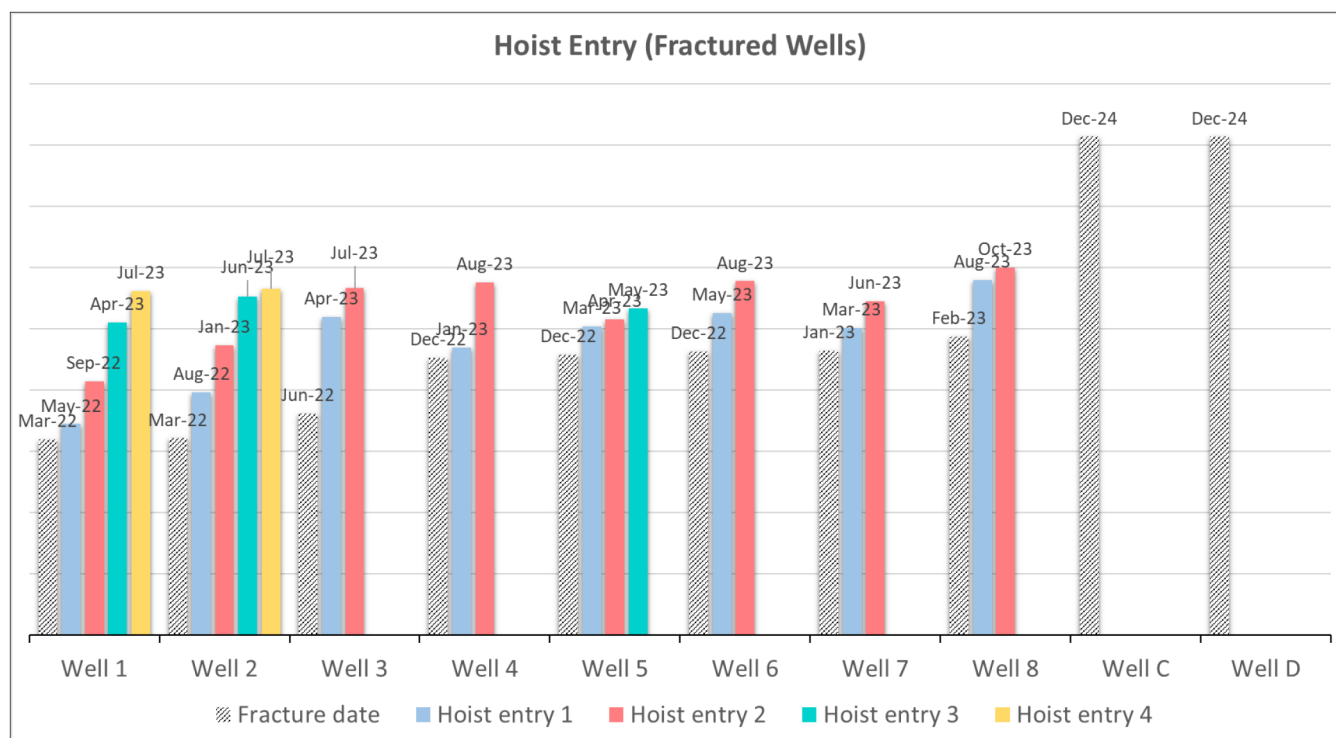


Figure 16—Lifetime intervention well comparison.

This reduction in intervention frequency not only minimized operational disruptions but also resulted in cost savings and improved overall well performance.

The long-term production performance and enhanced recovery results from these wells will be discussed in detail in a separate section.

Conclusion and Way Forward for Future Implementation

This paper presents the first successful deployment of rod-shaped proppant in the Area-1 field, marking a significant milestone in the region's hydraulic fracturing operations. The data collected during this initial trial offers valuable insights and establishes a foundation for future development and optimization efforts. The key lessons learned from this pioneering implementation are summarized below:

- **Operational and Production Benefits.** The successful application of the new perforation strategy combined with rod-shaped proppant in Wells C and D demonstrated clear operational and production advantages. These included the absence of proppant flowback, extended operational life without interventions, and enhanced production performance. Insights from long-term monitoring will guide future fracturing designs and contribute to continuous improvement in well performance and operational efficiency.
- **High Concentration Compatibility.** The use of rod-shaped proppant at concentrations up to 7 PPA did not result in any operational issues when paired with the larger Entrance Hole Diameter (EHD). This suggests that higher concentrations can be safely considered in future applications.
- **Flexible Deployment Options.** Future implementations may explore increasing the volume of rod-shaped proppant to further extend well life and reduce flowback risks.
- **Frac Equipment Compatibility.** Rod-shaped proppant is fully compatible with standard well-site frac equipment, requiring no modifications for delivery or handling, which simplifies logistics and reduces deployment costs.

- Operational Best Practices. Adherence to standard best practices is essential, particularly during the transition phase when switching from conventional to rod-shaped proppant—especially when used as a tail-in. Proper handling and monitoring during this stage help ensure consistent proppant concentration and stable pressure trends.

Acknowledgements

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